CALL MANAGEMENT CRM INTEGRATION

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Executive Summary

Capital Stack Operations Agent: An AI-Powered Automation System

Background

Capital Stack is a pay-per-call marketing firm that supports investment companies operating under SEC Regulation 506(c). In recent years, the company has experienced significant growth; however, its existing systems have created a capacity limit that prevents it from taking on additional clients. Manual CRM updates, high-volume communication tasks, and multi-step billing processes restricted operational efficiency and capped monthly revenue at approximately $60,000. Although the demand for Capital Stack's services continues to rise, the current platforms do not provide the level of automation, accuracy, or scalability required to support expansion.

Project Overview

The Capital Stack Operations Agent is an AI-driven automation system designed to streamline core operational activities and reduce manual workload. This platform integrates real-time communication, CRM management, workflow automation, and billing execution into one unified system. It connects directly with Wavv Dialer, GoHighLevel, Slack, and Stripe to update lead records, analyze call transcripts, send appointment confirmations, execute billing processes, and deliver operational reports.

This project outlines the development of the Capital Stack Operations Agent by applying systems analysis and design tools throughout the design process. UML diagrams, such as Activity Diagrams, Use Case Diagrams, Sequence Diagrams, and Class Diagrams, were used to demonstrate system functionality, user interactions, and workflow behavior. Additionally, analysis documents defining both the functional and non-functional requirements of the system were developed to ensure that all operational needs are met. Visual materials were also developed to illustrate the user interface and workflow structure, providing a clear and comprehensive view of the proposed system.

Conclusion

The solutions provided by the Capital Stack Operations Agent increase efficiency, improve accuracy, and support higher call and client volumes. The system positions the company for significant operational improvement while maintaining service quality and reliability. Over time, this platform will become an essential tool for Capital Stack by enabling scalable growth, strengthening client relationships, and enhancing overall business performance.

System Request

PROJECT SPONSOR

Tharun Senthil

BUSINESS NEED

Capital Stack, a firm that provides pay-per-call marketing services for investment firms operating under SEC Regulation 506(c), is currently facing a capacity constraint. The team cannot take on additional clients because they have reached the maximum number of calls they can manage with existing systems. This constraint prevents the company from expanding its operations and keeps revenue capped at approximately $60,000 monthly. Implementing a new system would allow Capital Stack to increase its operational capacity, attract a larger client base, and capitalize on new business opportunities.

BUSINESS REQUIREMENTS

The new system is expected to fulfill the following business requirements:

- Automate repetitive CRM, messaging, reporting, and billing tasks to reduce manual effort and improve efficiency.
- Use call transcript analysis to update CRM records, send appointment confirmations, and manage daily call lists dynamically.
- Centralize client dialing and messaging under one account for streamlined, consistent operations.
- Integrate with Slack and GoHighLevel to update internal tracking and execute billing automatically in real time.
- Include human oversight so SDRs can review, confirm, or override automated system actions.

BUSINESS VALUE

In the short term, this new system would raise Capital Stack's lead capacity from 1,500 - 2,000 to 3,000 - 4,000 per month, enabling the company to increase its revenue to $90,000 - $120,000 per month while maintaining the same team size. In the long run, higher margins would make it financially feasible to hire experienced sales and operations professionals, leading to improved performance, strengthened client satisfaction, and sustained growth.

SPECIAL ISSUES OR CONSTRAINTS

- Must integrate with existing CRM and systems without infrastructure replacement
- Reliance on third-party software may cause operational disruptions if pricing, features, or availability change
- Must be built and tested within one month to avoid extra costs from the $500 monthly Wavv Dialer subscription

Feasibility Analysis

Capital Stack is developing an AI-powered operations automation system that integrates with Wavv Dialer, GoHighLevel (CRM), and Slack to streamline internal processes and reduce manual workload. This section provides a detailed evaluation of the technical feasibility of the proposed system. The purpose of this solution is to enhance efficiency, minimize repetitive operational tasks, and increase revenue by improving scalability and productivity across the business.

TECHNICAL FEASIBILITY

The project is feasible due to the team's possession of the essential technical skills, tools, and system knowledge required for implementation. The team has prior experience working with CRM platforms, workflow automation, and application integration, which reduces development risk and ensures an understanding of the required functionality.

From a technology perspective, the system is achievable because Wavv Dialer now offers API access, enabling integration with other platforms such as GoHighLevel and Slack. However, the solution relies on multiple third-party platforms, which may introduce limitations in configuration or data synchronization. This dependency on external APIs presents integration challenges that must be managed carefully to avoid system disruptions.

In terms of size, the project is moderate in complexity. It will be developed by a small team, which limits available manpower and increases individual responsibility for development and testing. This small team size introduces some risk, as unexpected technical challenges may delay progress. The estimated development timeline is approximately one month, leaving minimal flexibility for unplanned revisions. The scope of work includes building AI-driven automation workflows, enabling secure API connections, and ensuring accurate synchronization of client records, messages, and reports.

The most significant technical risk involves ensuring seamless integration among Wavv Dialer, GoHighLevel, and Slack. Potential issues include inconsistent API behavior or delays in real-time data updates, which may affect operational reliability. Nevertheless, the team's technical competency and access to integration tools provide confidence that these challenges can be addressed with proper planning and testing.

Overall, the system is technically feasible, though careful management of integration dependencies and testing will be essential to ensure long-term stability and scalability.

ECONOMIC FEASIBILITY

The financial justification for this project is solid. A comprehensive cost–benefit analysis was performed to assess both tangible and intangible elements and determine the project's return on investment (ROI), net present value (NPV), and break-even point.

Costs and Benefits

The evaluation considered development costs, operational costs, tangible benefits, and intangible benefits. Development costs included system setup and testing, consulting fees, data conversion, and training expenses. Operational costs included recurring subscription charges, software upgrades, maintenance, and communication costs. In regard to tangible benefits, the project is anticipated to yield increased sales revenue, reduce staffing requirements, and enhance call efficiency. Intangible benefits include improved customer satisfaction, stronger brand reputation, improved client relationships, and raised employee morale.

Development Costs	Operational Costs
Software Installation	Ongoing Subscription Fees
System Configuration	Software Upgrades
Consulting Fees	Routine Maintenance
Development Training	Outgoing Call Fees
System Testing	Communication Costs
Data Conversion Costs	User Training
Tangible Benefits	**Intangible Benefits**
Higher Sales Revenue	Higher Customer Satisfaction
Reduced Staffing Needs	Stronger Brand Reputation
Improved Call Efficiency	Expanded Market Presence
Increased Client Onboarding	Improved Employee Morale
Lower Hiring Costs	Better Client Relations

Assigning Values to Costs and Benefits

The chart on the right shows the breakdown of values. Increased sales were estimated at $330,000 annually, improved call efficiency at $50,000, and reduced staffing costs at $120,000, resulting in total annual tangible benefits of $500,000. Development costs, including consulting, software, and developer labor, totaled $5,900, while recurring operational costs (software, maintenance, and subscription fees) added up to $16,000. This results in total costs of $21,900 in Year 1.

Benefits	
Increased Sales	$330,000
Improved Call Efficiency	$50,000
Reduced Staffing Costs	$120,000
Total Benefits	**$500,000**
Development Costs	
Consulting Fees	$200
Server Software	$500
Development Labor	$5,000
System Testing	$200
Total Development Costs	**$5,900**
Operational Costs	
Software	$4,800
Routine Maintenance	$5,200
Call Subscription	$6,000
Total Operational Costs	**$16,000**
Total Costs	**$21,900**

Cash Flows and Net Present Value (NPV)

	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Increased Sales	$330,000	$360,000	$360,000	$360,000	$360,000	
Improved Call Efficiency	$50,000	$50,000	$50,000	$50,000	$50,000	
Reduced Staffing Costs	$120,000	$120,000	$120,000	$120,000	$120,000	
TOTAL BENEFITS:	$500,000	$530,000	$530,000	$530,000	$530,000	
PV OF BENEFITS:	**500,000**	**490,741**	**454,389**	**420,726**	**389,560**	**2,255,416**
PV OF ALL BENEFITS:	**500,000**	**990,741**	**1,445,130**	**1,865,856**	**2,255,416**	
Consulting Fees	200	0	0	0	0	
Server Software	500	0	0	0	0	
Development Labor	5,000	0	0	0	0	
System Testing	200	0	0	0	0	
TOTAL DEVELOPMENT COSTS	5,900	0	0	0	0	
Software	4,800	4,800	4,800	4,800	4,800	
Routine Maintenance	5,200	5,200	5,200	5,200	5,200	
Call Subscription	6,000	6,000	6,000	6,000	6,000	
TOTAL OPERATIONAL COSTS:	16,000	16,000	16,000	16,000	16,000	
TOTAL COSTS:	21,900	21,900	21,900	21,900	21,900	
PV OF COSTS:	**21,900**	**20,278**	**18,774**	**17,379**	**16,093**	**94,424**
PV OF ALL COSTS:	**21,900**	**42,178**	**60,952**	**78,331**	**94,424**	
TOTAL PROJECT BENEFITS - COSTS:	**478,100**	**508,100**	**508,100**	**508,100**	**508,100**	
YEARLY NPV:	**478,100**	**470,463**	**435,615**	**403,347**	**373,467**	**2,160,992**
CUMULATIVE NPV:	**478,100**	**948,563**	**1,384,178**	**1,787,525**	**2,160,992**	
RETURN ON INVESTMENT:	**22.89%**	(2,160,992/94,424)				
BREAK-EVEN POINT:	**.06 years**	[break-even occurs in Year 1]				
INTANGIBLE BENEFITS:	Improved efficiency, customer satisfaction, brand reputation, staff morale					

The chart above indicates cash flows projected over a five-year period. Total benefits increase from $500,000 in Year 1 to $530,000 by Year 5, resulting in a cumulative benefit of $2.55 million. The present value of these benefits, discounted at 8%, amounts to $2.26 million, while the present value of costs is only $94,424. Consequently, the cumulative NPV over five years is around $2.16 million, indicating a positive return.

Return on Investment (ROI)

The ROI for the first year was calculated at 22.89%, with returns significantly increasing in subsequent years as benefits accumulate and costs remain low.

Break-Even Analysis

The break-even point is reached almost immediately, within 0.06 years (less than the first month), indicating that the project recovers its investment costs right at the start of Year 1. This swift payback underscores the strength of the financial case.

Due to the minimal initial costs, low ongoing expenses, and very high anticipated benefits, the project is economically viable. The strong ROI, immediate break-even point, and high cumulative NPV illustrate that this project recoups its costs quickly and generates considerable long-term financial value.

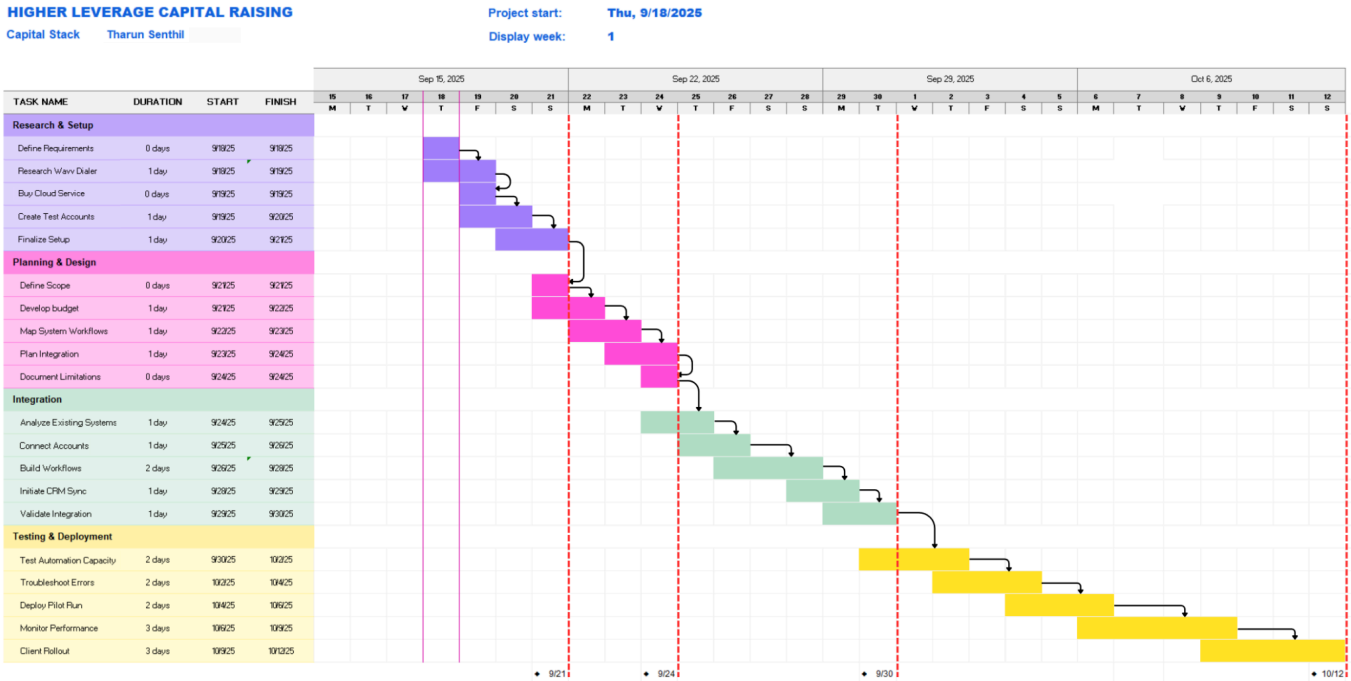
ORGANIZATIONAL FEASIBILITY

The organizational feasibility of this project is strong, as it directly supports Capital Stack's strategic objective of scaling operations without increasing labor costs. By automating repetitive operational tasks such as CRM updates, client follow-ups, and reporting, the system strengthens the company's ability to manage higher lead volumes and onboard additional clients efficiently. Organizational support is high, as both the project sponsor and key team members recognize the significant time savings and long-term business value that automation will deliver.

For users, adoption is expected to be positive. The AI-driven system reduces manual workload and streamlines daily responsibilities, making it especially beneficial for operations and sales teams who currently spend substantial time on routine administrative tasks. While some training will be required to adapt to automated workflows and new integrations with Wavv Dialer, GoHighLevel, and Slack, the learning curve is manageable, and the transition is expected to be smooth. Any short-term adjustment will be outweighed by improved productivity and ease of use.

Other stakeholders, such as clients, will also benefit from faster communication, consistent follow-ups, and greater service reliability. These improvements enhance customer experience and strengthen long-term client relationships by projecting professionalism and responsiveness. Overall, strong leadership support, clear user benefits, and positive stakeholder impact ensure that the system is well-positioned for successful adoption across the organization.

Gantt Chart



Explanation

The Gantt chart outlines Capital Stack's project in four phases: Research & Setup, Planning & Design, Integration, and Testing & Deployment. Each phase includes specific tasks with defined start and finish dates, displayed in color-coordinated blocks for clarity. The diamonds represent milestones, which mark important checkpoints at the end of each phase, such as finalizing setup, documenting limitations, validating integration, and client rollout. Additionally, the arrows represent task dependencies, illustrating the sequential flow of the project where each task must be completed before the next one can begin. The first two weeks are centered around quick setup and preparation tasks. However, as the project continues, the focus shifts to more complex integration work and multi-day testing. Overall, the chart provides a clear roadmap that keeps the project organized, highlights key milestones, and ensures the timeline stays on track to finish within one month.

Requirements Definition

SYSTEM NAME

Capital Stack Operations Agent

PURPOSE

The Capital Stack Operations Agent is an AI-driven automation system designed to eliminate repetitive operational tasks performed by Capital Stack's team. The system integrates with Wavv Dialer, GoHighLevel CRM, and Slack to automate CRM updates, client messaging, billing actions, and operational reporting. Its purpose is to reduce manual workload, improve process efficiency, and increase scalability.

FUNCTIONAL REQUIREMENTS

FR-1: Real-Time and Scheduled Task Execution

The system must execute operational automations based on predefined timing categories. Real-time tasks should run immediately when triggered by events such as call completion or CRM updates. Morning workflow tasks should be executed before the business day begins to prepare daily reports, and end-of-day workflows should finalize logging, attendance validation, and billing processes.

FR-2: AI-Based Task Recognition

The system should use artificial intelligence (AI) and natural language processing (NLP) to interpret call transcripts, Slack messages, and webhook inputs. Based on this contextual analysis, the system will be able to automatically initiate workflows such as tagging leads, assigning statuses, updating notes, or generating follow-up actions.

FR-3: Automate CRM Management

The system must automatically update lead records within GoHighLevel by modifying fields such as lead stage, client status, notes, and interaction history. These updates will occur in response to either AI-recognized call outcomes or workflow triggers, eliminating the need for manual CRM entry.

FR-4: Automate Repetitive Client Messaging

The system must send automated client communications triggered by business events or scheduled workflows. This includes appointment confirmations, post-call follow-ups, reminders, no-show notifications, end-of-day summaries, and re-engagement

sequences. The system should determine message timing and content based on predefined logic or AI interpretation of context.

FR-5: Integrate with APIs

The system must securely connect to Wavv Dialer, GoHighLevel CRM, Slack, and Stripe through their respective APIs to exchange data, trigger automations, and execute actions programmatically. All integrations should utilize authenticated API connections to maintain data security.

FR-6: Human-in-the-Loop Interaction

The system must allow Sales Development Representatives (SDRs) to review, approve, or reject AI-generated actions directly through Slack. The system should interpret user feedback provided in natural language and implement human revisions before executing actions that require validation, such as client messages or billing-related notifications.

FR-7: Reporting and Transparency

The system must generate both real-time and scheduled operational reports summarizing tasks performed, pending actions, and exceptions. These reports should be delivered through Slack for team visibility and may include lead status updates, failed task alerts, and process summaries.

FR-8: Error Handling and Task Validation

The system must detect failed automations, API errors, or incomplete workflows and notify the assigned SDR for manual review. The system should maintain detailed logs of all automation events, execution statuses, and error messages to support traceability and auditing.

FR-9: Automate Billing

The system must automatically identify billable events such as qualified calls, attended appointments, or fulfillment milestones, and initiate daily billing operations through Stripe. The system should also record and store billing history and payment status in GoHighLevel or internal tracking sheets for revenue reporting.

NONFUNCTIONAL REQUIREMENTS

Operational Requirements

NFR-1.1: The system should operate primarily through Slack as the central interface for workflow approvals, status updates, and AI reporting.

NFR-1.2: Integrations with GoHighLevel, Stripe, Google Sheets, and Slack must function without disrupting existing business workflows.

NFR-1.3: The system should operate as an internal asset managed entirely by Capital Stack, without reliance on any external or third-party automation services.

NFR-1.4: All reporting, notifications, and monitoring activity must be delivered through Slack to maintain consistent operational oversight.

Performance Requirements

NFR-2.1: Real-time automations triggered by system events must be executed and completed within 3–5 minutes.

NFR-2.2: Scheduled automations, such as morning summaries, must be completed by 8:05 a.m. daily.

NFR-2.3: Pre-meeting reminders must be delivered exactly one hour before the scheduled appointment time.

NFR-2.4: End-of-day workflows must be completed within 5 minutes of their trigger time.

NFR-2.5: Data synchronization and API-based updates must be processed with minimal latency.

Security Requirements

NFR-3.1: All system code and automation logic should be hosted securely within Capital Stack's private Google Cloud Platform (GCP).

NFR-3.2: No client or automation data should be stored in unauthorized third-party tools.

NFR-3.3: All credentials, API keys, and webhooks must be encrypted and stored securely within Capital Stack's infrastructure.

NFR-3.4: GoHighLevel should not store any automation credentials due to client visibility in sub-accounts.

NFR-3.5: All API integrations should use role-based access controls (RBAC) and limited-scope API keys.

Cultural and Political Requirements

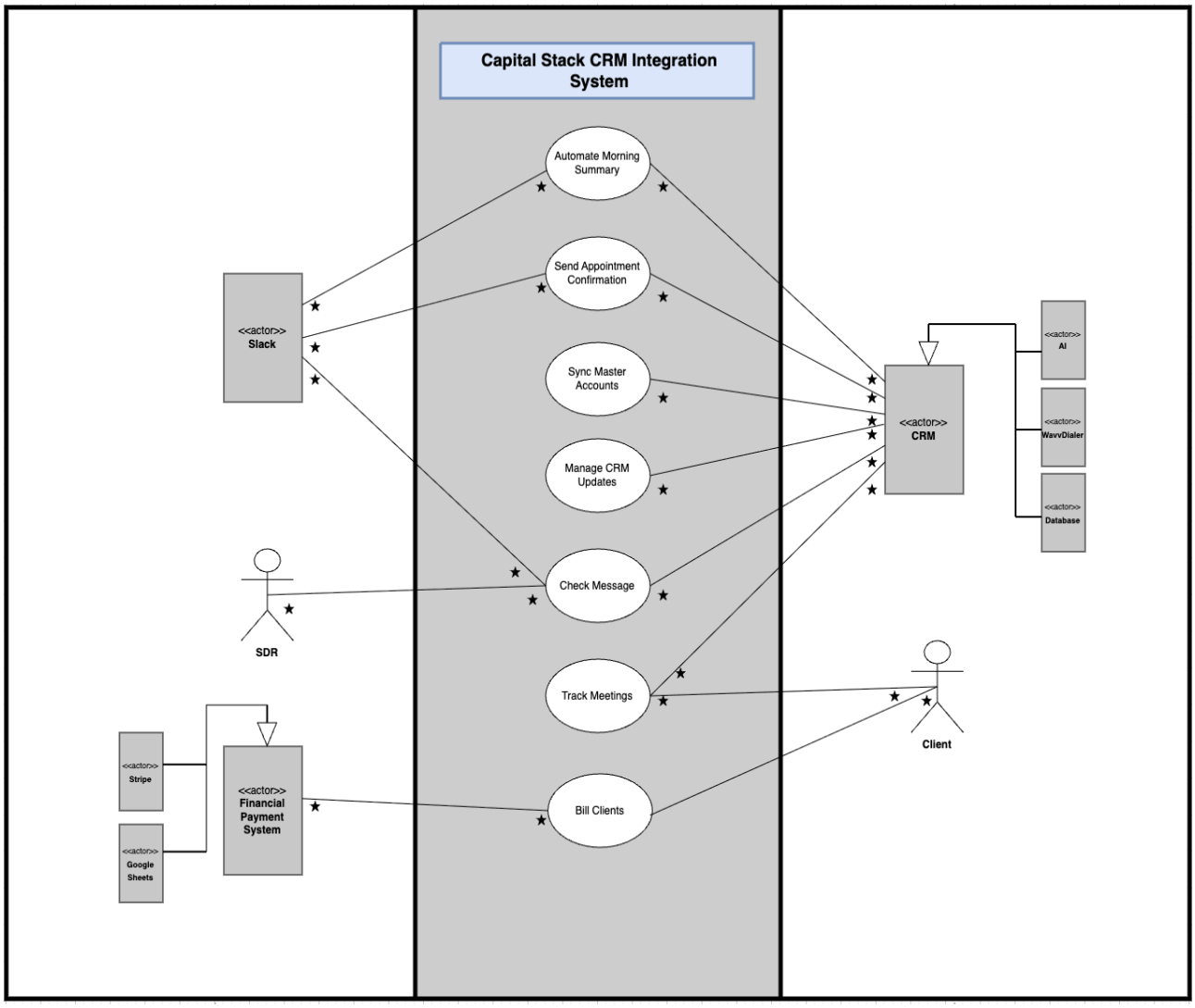
NFR-4.1: Automated communications must align with Capital Stack's professional tone, client communication standards, and brand guidelines.

NFR-4.2: The system must operate solely on Capital Stack's approved data sources, prohibiting the use of unverified or third-party data sources.

NFR-4.3: AI-generated content must maintain transparency, accuracy, and professionalism.

NFR-4.4: The system should uphold company values regarding client trust, data integrity, and ethical automation.

Use Case Diagram



Use Case Descriptions

Use Case Name: Send Appointment Confirmation	**ID:** 2	**Importance Level:** High
Primary Actor: SDR (Sales Development Representative)	**Use Case Type:** Detail, Essential	

Stakeholders and Interests:
SDR— wants accurate, compliant messages sent to clients with minimal rework.
Client— wants a timely, clear confirmation of their scheduled appointment.

Brief Description: This use case describes how the system processes a recorded call, verifies client information, generates an appointment confirmation message, and ensures SDR review before sending the message to the client.

Trigger: A client call is recorded and logged into the CRM system

Type: External

Relationships:
Association: Slack, AI, Wavv Dialer, Database
Include:
Exclude:
Generalization:

Normal Flow of Events:
1. Wavv Dialer records a call from the client and syncs the call log to the CRM.
2. The Database retrieves the client's data.
3. The AI System confirms the Client's identity using fetched data.
4. The AI System analyzes the call transcript and generates a call summary.
5. The Database updates the CRM with the new client information.
6. The Database checks if the Client has an existing appointment.
   If the client has an existing appointment,
   the S-1: appointment confirmation subflow is performed.
7. The SDR reviews the appointment confirmation message.
   If the SDR approves the message,
   the S-2: message approval subflow is performed.
8. The Client receives the appointment confirmation message.

SubFlows:
S-1: Appointment Confirmation
1. The AI system generates a draft confirmation message.
2. The AI system notifies the SDR that the message is ready for review.

S-2: Message Approval
1. The SDR approves the confirmation message.
2. The SDR sends the confirmation to the client.

Alternate/Exceptional Flows:
S-1, 1a1: When no appointment exists, the system saves the call summary to the CRM and terminates the process without generating a confirmation message.
S-2, 1a1: The SDR rejects the confirmation message, the AI system routes the draft for revision and resubmits the updated message to the SDR for review.

Use Case Name: Bill Clients	**ID:** 7	**Importance Level:** High
Primary Actor: SDR (Sales Development Representative)	**Use Case Type:** Detail, Essential	

Stakeholders and Interests:
SDR — wants accurate billing records sent to clients with minimal manual effort and proper approval workflow.
Client — wants to receive accurate, timely invoices for services rendered.

Brief Description: This use case describes how the system processes attendance records, generates billing information, creates billing summary reports, manages approval workflows, and sends invoices to clients while logging payment records..

Trigger: End of workday is at 8 pm.

Type: External

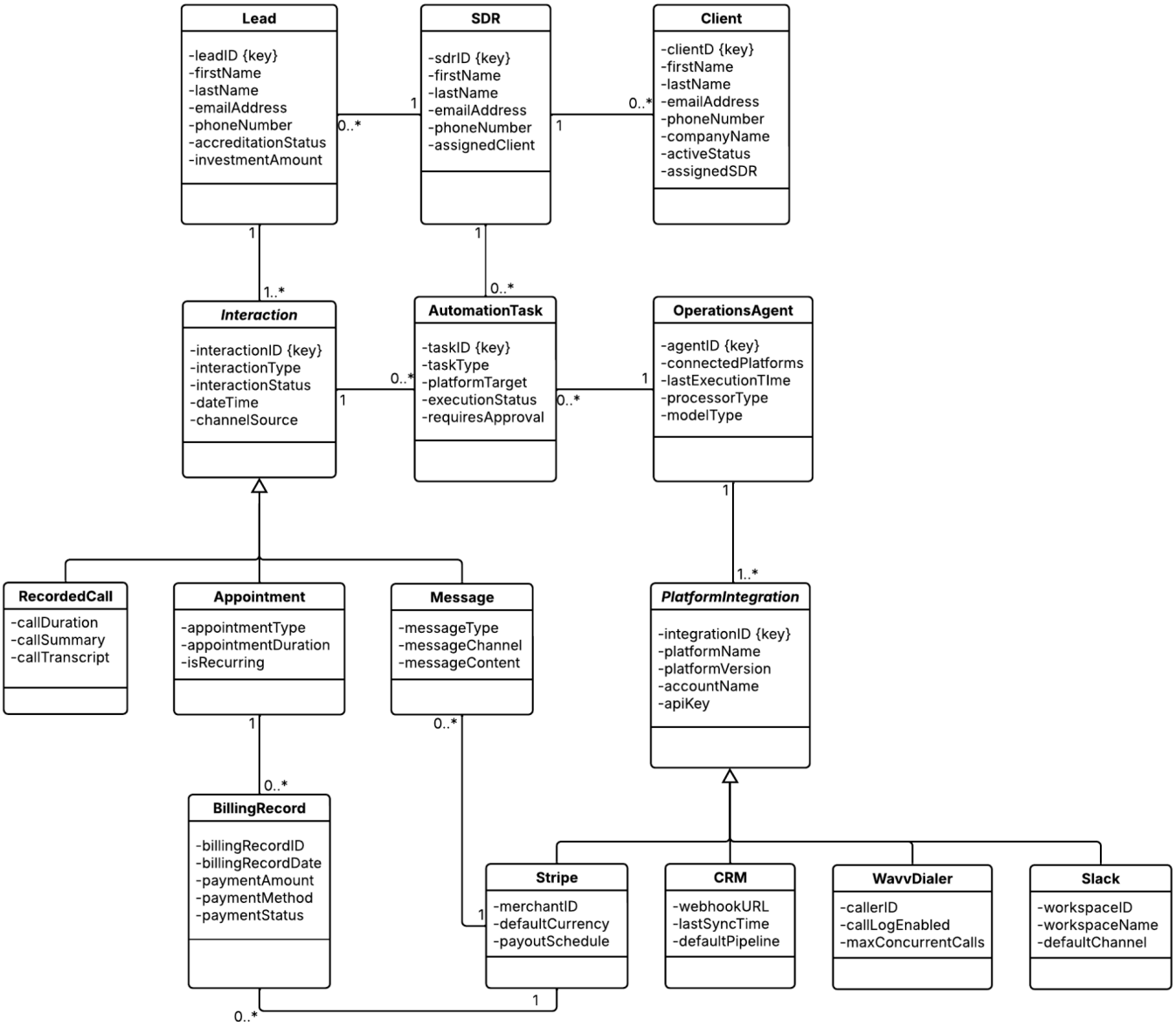
Relationships:
Association: GoHighLevel, Google Sheets, AI, SDR, Stripe

Normal Flow of Events:
1. The system identifies the end of the workday at 8 pm.
2. GoHighLevel fetches attendance data and sends it to the AI system.
3. The AI system compares data with the Google Sheets tracker.
4. The AI System determines who showed up and who didn't.
5. GoHighLevel updates the attendance data from the AI system.
6. Google Sheets is used to log billing records.
7. The AI System creates a summary report and sends the report to the SDR.
8. The SDR reviews the billing information.
   If the SDR approves the billing,
   the S-1: approved billing subflow is performed.
9. Google Sheets logs the payment.

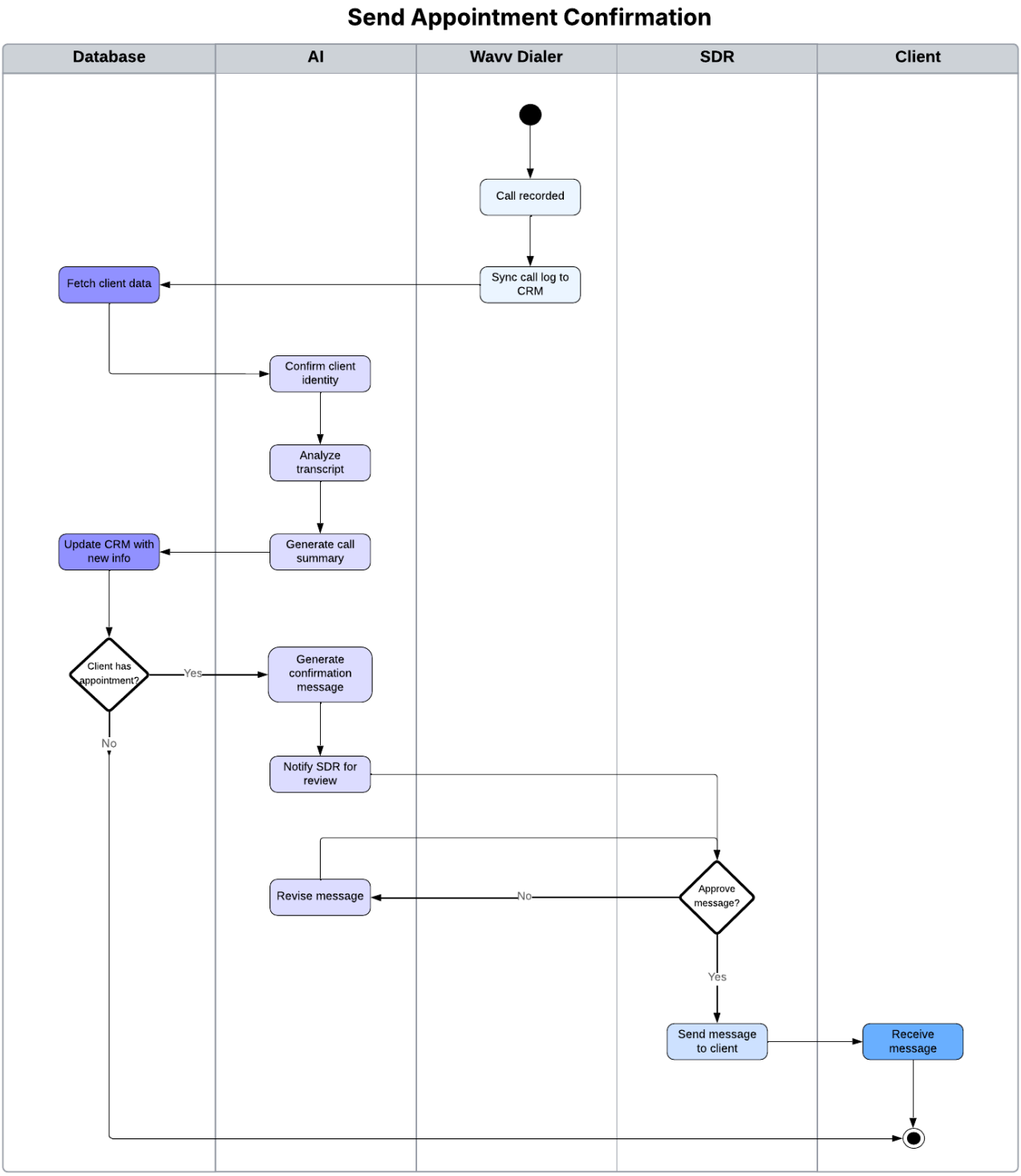
SubFlows:
S-1: Approved Billing
1. The SDR approves the billing in the system
2. Stripe bills or requests payment from the client.
3. Stripe confirms that the payment is received.

Alternate/Exceptional Flows:
S-1, 1a1: When the SDR rejects the billing, the billing records are manually revised by the SDR and sent to Google Sheets to update the billing record before routing to Stripe to bill the client.

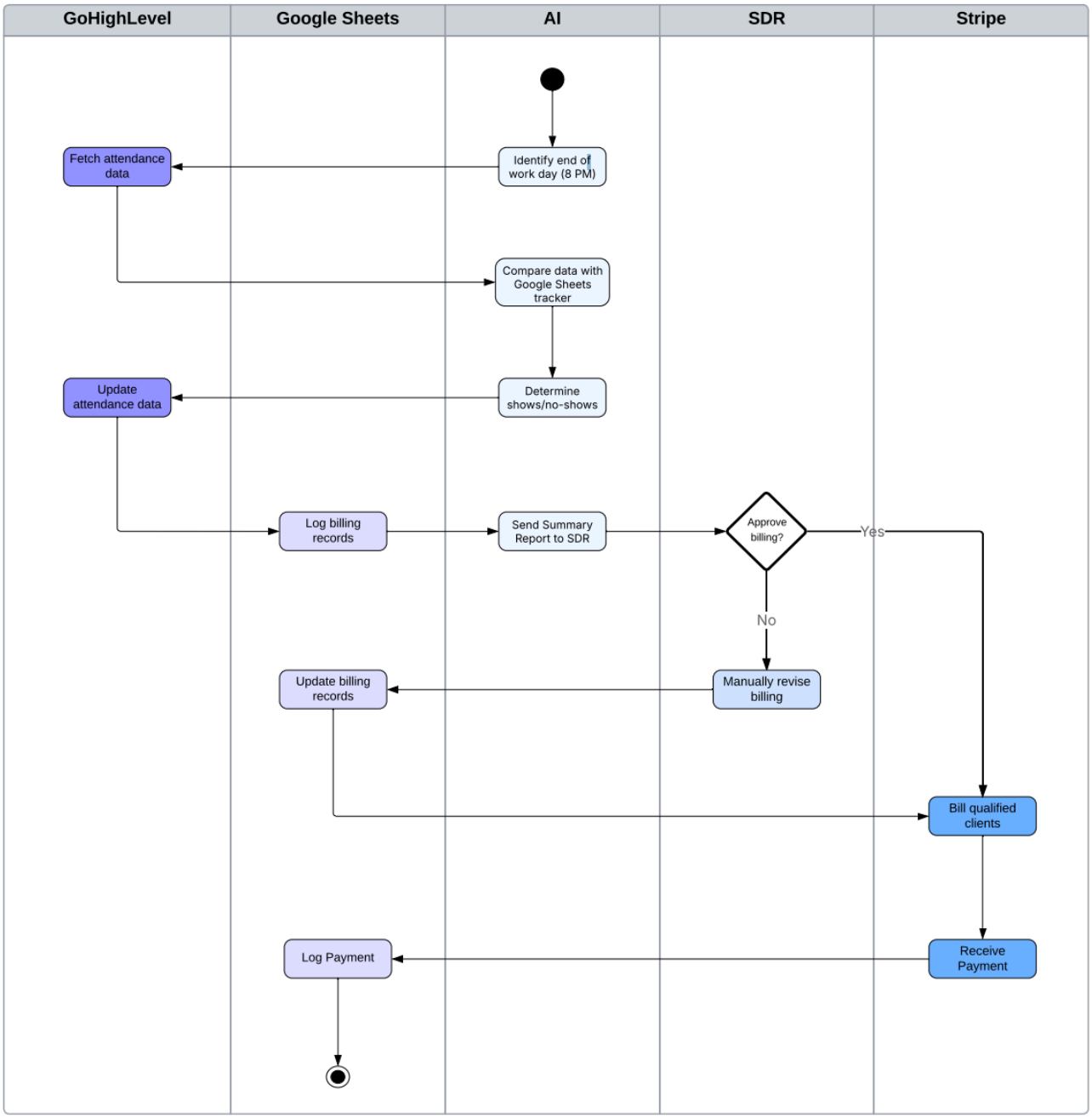
Domain Model Class Diagram



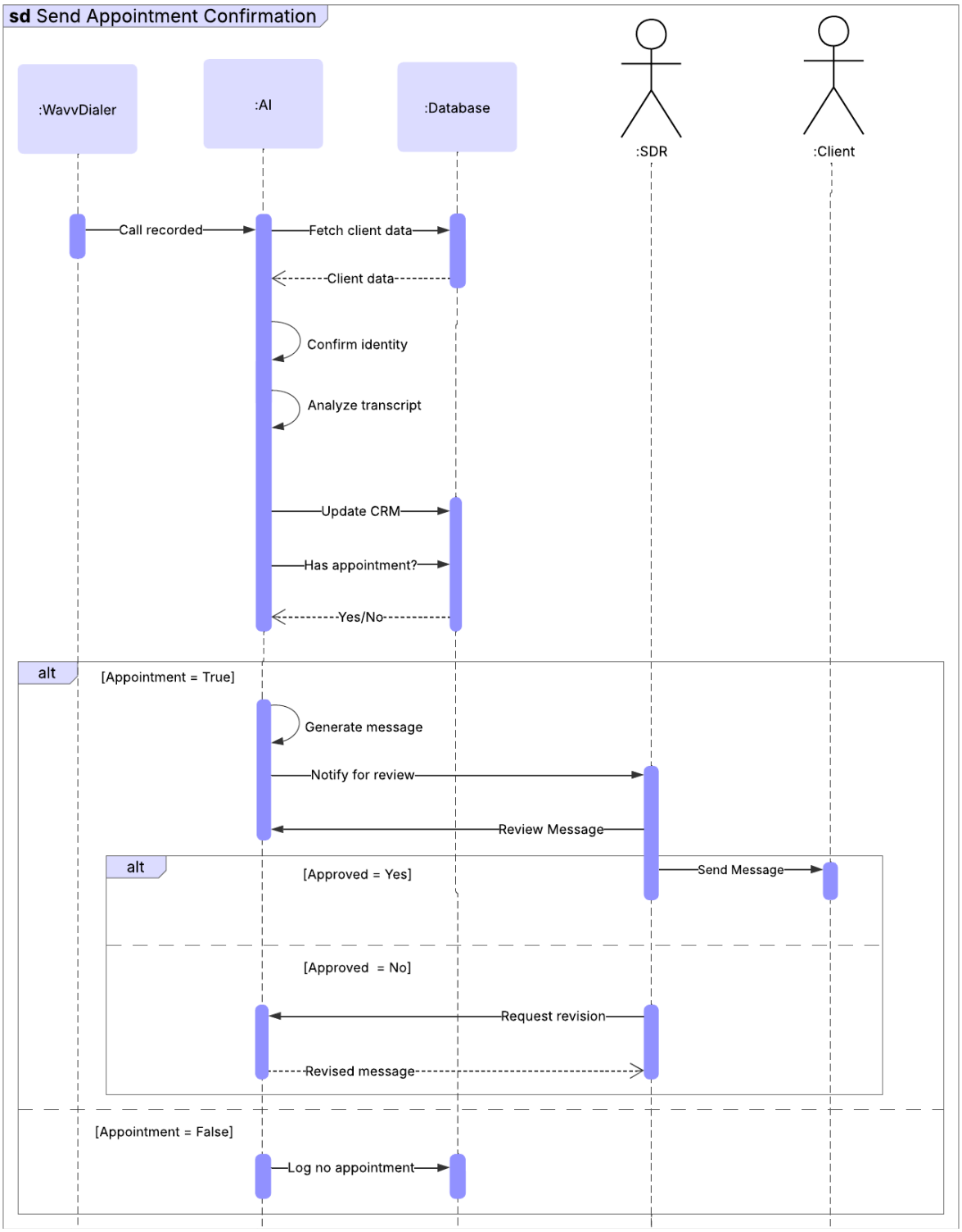
Activity Diagrams



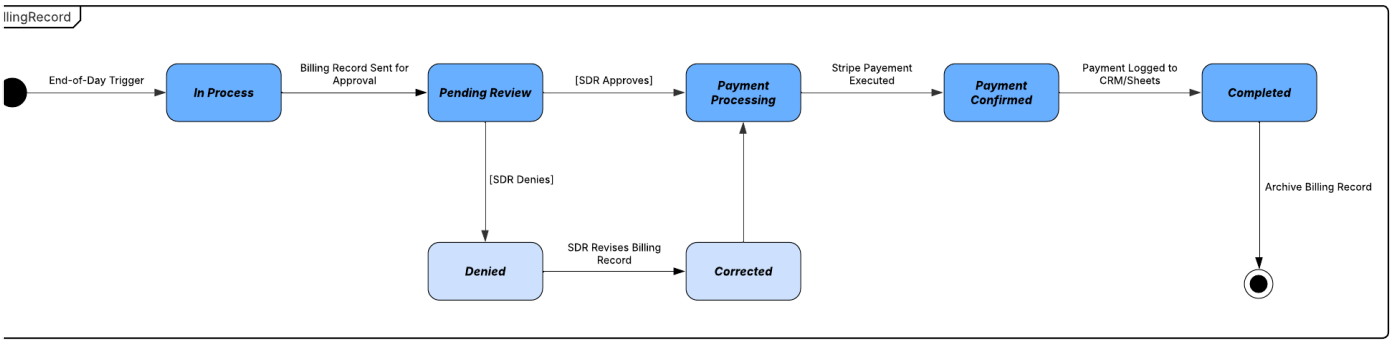
Bill Clients



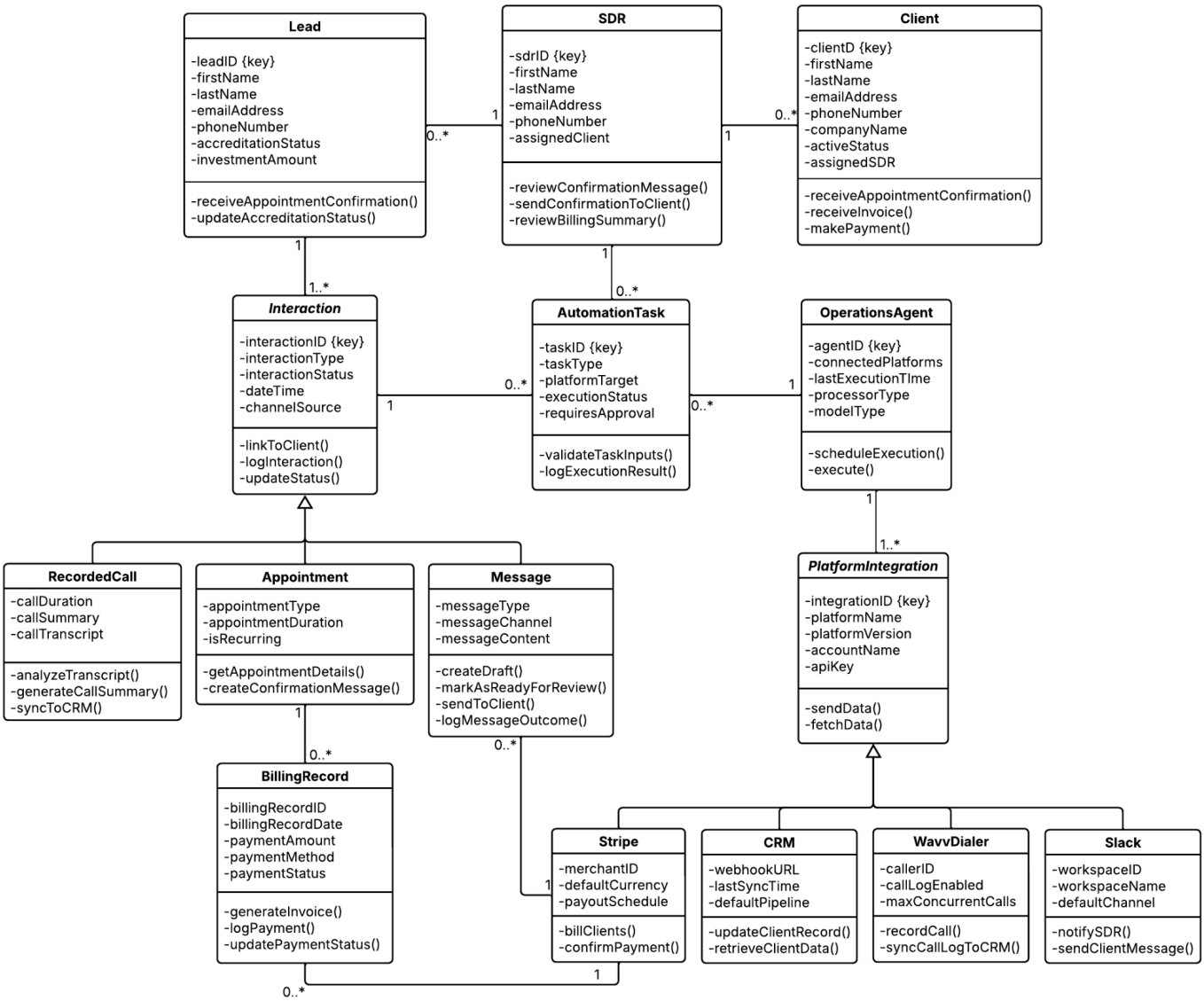
System Sequence Diagram



State Machine Diagram



Design Class Diagram

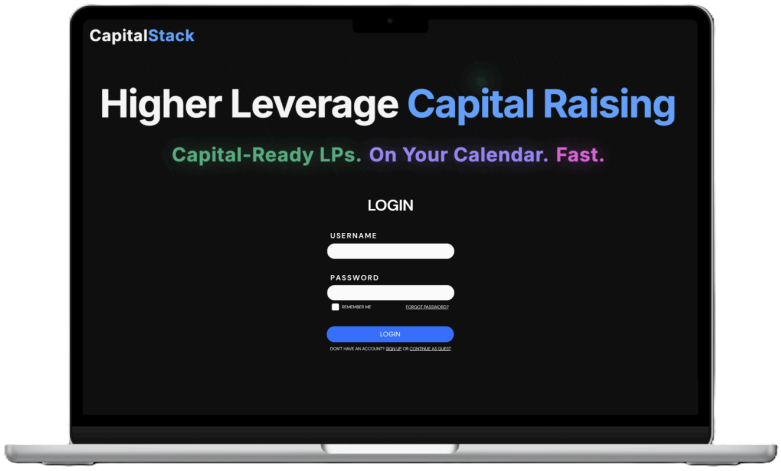


User Interface Design

Introduction

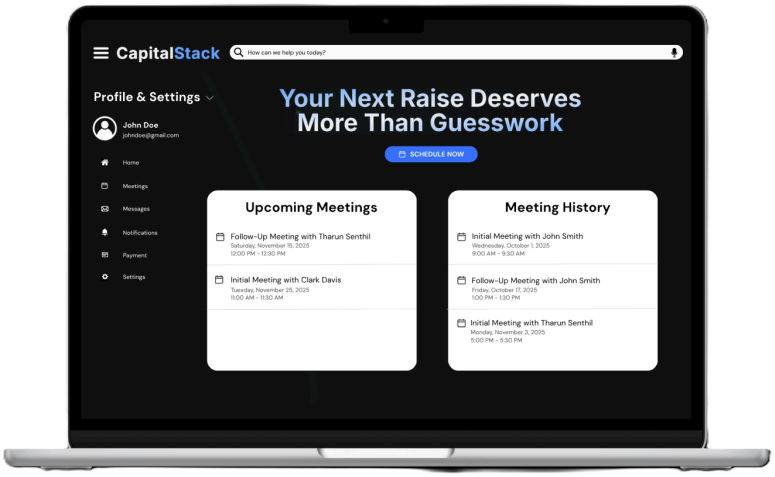
The following UI storyboard illustrates how the Sales Development Representative (SDR) uses Capital Stack's platform to schedule a meeting for a lead. Each screen represents a step in the process, starting from logging in and navigating the dashboard to selecting a meeting time, entering lead information, and viewing the final confirmation. The goal of this storyboard is to clearly demonstrate how the system supports the SDR's workflow through simple and organized user interfaces.

Login and Welcome Screen



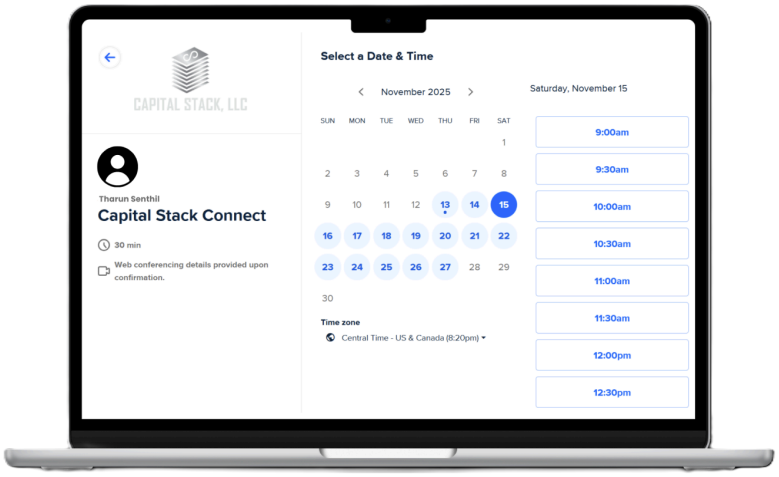
The first screen welcomes the SDR to the Capital Stack platform. At this stage, the SDR logs in by entering their username and password and clicking the Login button. Additional interactive elements include "Forgot Password?", "Sign Up", and a "Remember Me" checkbox, which enhance usability and account recovery options.

Home Dashboard and Navigation



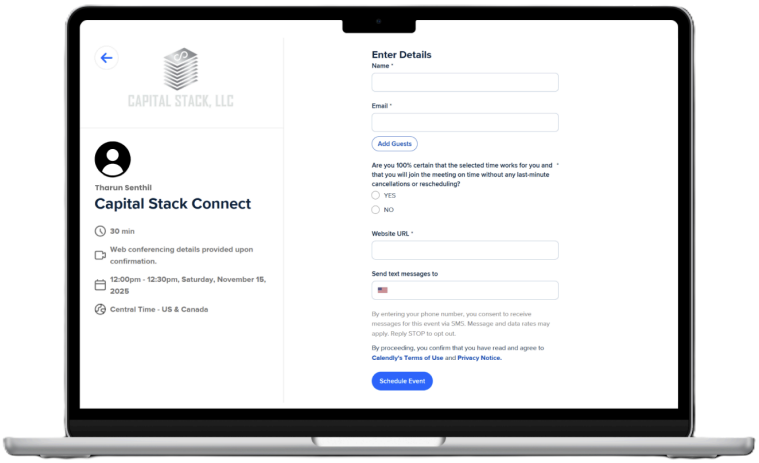
After the SDR successfully logs in, they are directed to the home screen, which serves as the primary navigation area. A hamburger menu at the top left expands into the Profile & Settings panel, providing access to key options, including Home, Meetings, Messages, Notifications, Payments, and Settings. This storyboard focuses on the Meetings section, where the SDR can view upcoming meetings and the lead's meeting history. The "Schedule Now" button, located at the top, allows the SDR to begin scheduling a new meeting.

Date and Time Selection Screen



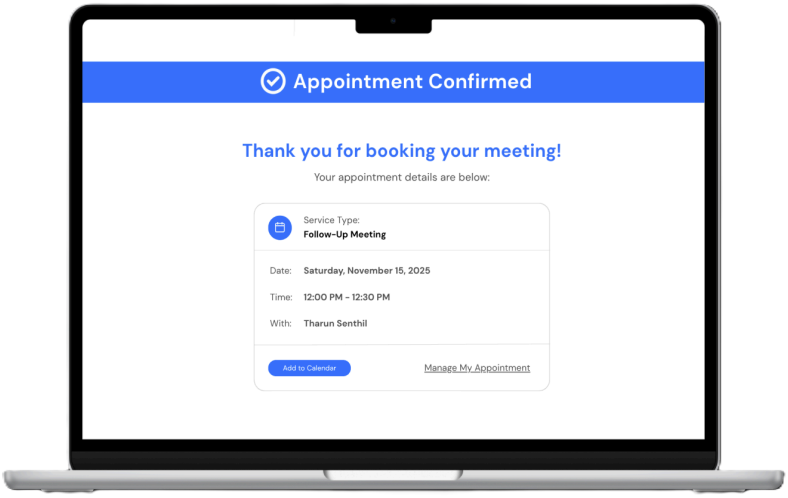
After selecting "Schedule Now," the SDR is taken to the scheduling screen. The left side of the screen shows basic meeting information, while the center displays an interactive calendar where the SDR selects an available date. After choosing the date, the SDR picks a time slot from the options listed on the right side of the screen. Once selected, the chosen date and time are highlighted in a darker shade of blue.

Lead Information and Event Details Screen



The next screen collects the lead's meeting details. The SDR fills in required fields such as the lead's name, email, website, guest information, and text message preferences. The system also asks the SDR to confirm that the selected meeting time works well for the lead. On the left side, a summary of the meeting details, including the date and time, is shown. To complete the process, the SDR selects the "Schedule Event" button.

Appointment Confirmation Screen



The final screen provides a clear confirmation that the meeting has been successfully scheduled. It presents key appointment details, including the service type, date, time, and host. The SDR can take additional actions using the "Add to Calendar" button or the "Manage My Appointment" option. Simultaneously, Capital Stack's Operations Agent automatically sends a confirmation email to all participants.

Testing Plan

Testing Approach

Our testing approach for Capital Stack focuses on evaluating whether the system's core workflows function correctly and satisfy both the functional and non-functional requirements of the company. For integration testing, we will apply use case testing to confirm that key processes operate as intended. This includes verifying that appointment confirmations are sent with the proper message content and ensuring that the billing flow accurately calculates charges and requests SDR approval before processing. In addition to use case testing, we will conduct system interface testing to assess how well Capital Stack exchanges data with external platforms. These platforms include GoHighLevel, which stores and updates lead and appointment information, Slack, which sends appointment confirmation messages, and Stripe, which processes all financial transactions.

For system testing, we will focus on evaluating usability and performance to ensure the system supports daily operations. Usability testing will determine whether SDRs can navigate the system efficiently, understand screen layouts, and complete necessary scheduling and billing tasks without difficulty. This verifies that the system is user-friendly, allowing SDRs to complete their work without unnecessary delays. Performance testing will assess how the system manages heavy workloads, such as multiple SDRs scheduling appointments or updating client information simultaneously, while maintaining consistent response times and overall stability.

Overall, this testing plan provides a clear framework for evaluating the accuracy, reliability, and usability of the Capital Stack system. By addressing both functional and non-functional requirements, the approach ensures that the system design supports operational needs and delivers a consistent and reliable experience for users.

Test Cases

Test Case 1 – Send Appointment Confirmation

Test Case #: 1.1	**Use Case Name:** Send Appointment Confirmation
System: Capital Stack Operations Agent	**Subsystem:** Slack Bot (Automation)
Designed by: Capital Stack	**Design Date:** 12/07/2025
Executed by: Capital Stack	**Execution Date:** N/A
Short Description: Validate the appointment confirmation workflow	
Test Type: Integration Test	

Pre-conditions
A scheduled appointment exists in the GoHighLevel CRM with a valid meeting time
The system is fully functional and can trigger events 1 hour before the meeting
Slack bot is active in the appropriate channel
Prospect and client data are stored in the GoHighLevel CRM

Step	Action	Expected System Response	Pass/ Fail	Comment
1	System reaches 1 hour before appointment	The bot sends a Slack message including the prosect name, meeting time, client name, GoHighLevel URL (hyperlink), and hyperlinked phone number.	N/A	N/A
2	User replies with a numerical response (e.g., "1")	The bot returns a formatted confirmation message: "$ProspectName$ at $Time$ confirmed." Bot includes notes, phone number, email, and LinkedIn if available.	N/A	N/A
3	User reacts to the message with a check mark emoji	The bot automatically sends the finalized confirmation message into the appropriate client channel.	N/A	N/A

Post-conditions
1. Confirmation status is logged internally for reporting.

Test Case 2 – Bill Clients

Test Case #: 1.2	**Use Case Name:** Bill Clients
System: Capital Stack Operations Agent	**Subsystem:** Slack Bot (Automation)
Designed by: Capital Stack	**Design Date:** 12/07/2025
Executed by: Capital Stack	**Execution Date:** N/A
Short Description: Validate pre-billing verification and execution	
Test Type: Integration Test	

Pre-conditions
Billing data is available for the client in the billing system
Slack bot is active and /billing command is enabled.
User has permission to trigger billing actions.

Step	Action	Expected System Response	Pass/ Fail	Comment
1	User types /billing in Slack	The bot returns a pre-billing report showing all billable items for the client and a message: "Please verify."	N/A	N/A
2	User reacts to the report with a check mark emoji	The bot processes billing for all listed clients and charges them accordingly.	N/A	N/A
3	System completes billing	The bot sends a final report listing: (1) successfully billed clients and (2) any failed billings.	N/A	N/A

Post-conditions
1. Billing records are updated in the billing system.
2. Slack channel contains both the verification report and the final billing results.

Appendix

This appendix provides descriptions of the external platforms used within the Capital Stack automation system, including their role and functionality within operational workflows.

A.1 GoHighLevel (CRM)

GoHighLevel (GHL) is a customer relationship management platform designed for sales and service-based businesses. It centralizes client information, appointment scheduling, contact history, and communication tracking within a single system. It also offers API integration capabilities, which allow external systems to retrieve and update lead and appointment data in real time. Within Capital Stack, it functions as the primary data repository for leads and client activity, ensuring accuracy and consistency across automated workflows.

A.2 WavvDialer

WavvDialer is a cloud-based outbound dialing system used for managing sales calls and tracking engagement. It automatically logs calls, stores audio recordings, and generates transcripts that can be analyzed by external systems. When integrated with GoHighLevel, WavvDialer synchronizes call activity to client records, enabling automated follow-ups and call-based workflow triggers. Its purpose in the system is to provide accurate call data and support high-volume communication without manual intervention.

A.3 Slack

Slack is a business communication platform that enables structured team collaboration through real-time messaging. It supports workflow approvals, message routing, and notifications across teams. In this system, Slack functions as the human review checkpoint in automated processes. The AI agent sends billing summaries and client messages to Slack for team review and approval, ensuring quality control and risk mitigation while still maintaining the efficiency of automated operations.

A.4 Stripe

Stripe is a financial technology platform that enables businesses to process online payments and manage billing operations securely. It provides secure payment execution, transaction logging, customer billing profiles, and invoicing capabilities. Through Stripe's API, the system can automatically generate and process payments based on approved billing data while ensuring compliance and accurate transaction records. In addition, Stripe synchronizes payment status updates with internal records for financial reporting.